

Event Theme Based Outfit Recommendation System

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I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

.....
Signature of the supervisor:

.....
Date

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Signature of the co-supervisor:

.....
Date

ABSTRACT

In this research, a virtual assistant has been designed and implemented that uses AI-based methods to perform Event Theme Color Analysis and Outfit Recommendation. There is always an issue when trying to choose clothes at an event because users cannot find clothes that fit the theme color of the decorations used in event arrangements. Current systems for recommending fashionable clothing do not take into account any environmental information such as theme colors for events. The proposed solution can help people in selecting an appropriate dress for a particular event.

The novelty of the current solution is in considering event decorations and determining dominant colors that will guide the choice of clothing colors and styles. Computer vision techniques are used to analyze event decoration pictures uploaded to the system. Afterwards, an intelligent combination of occasion filter and fashion knowledge can be applied for generating a list of outfit ideas for men and women. The methodology starts with obtaining a collection of image datasets based on different occasions such as weddings, parties, corporate events, etc. After analyzing the images and determining the dominant colors, a mapping process is done based on color theory rules.

This system is deployed as a stack architecture that consists of a frontend interface developed using React and a backend server created using Flask. Users can upload images of the event, choose an occasion, and get outfit recommendations along with the recommended colors and style based on gender.

This study proves the effectiveness of incorporating the environment context in the fashion recommendation system. It provides a user with a decision-making aid when choosing the best fashion style that is visually matched for any occasion, but it is not meant to replace professional fashion consultation.

Keywords: Artificial Intelligence, Fashion Recommendation System, Computer Vision, Color Extraction, Event Theme Analysis, Color Theory, Context-Aware System.

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1. INTRODUCTION

Choosing appropriate clothes for different occasions is an essential element in personal presentation and visual aesthetics. In cases such as weddings, birthday parties, business events, and themed occasions, there is a necessity to choose clothes that complement not only the particular event but also the overall environment, including the décor and color schemes. However, choosing outfits to match event themes is rather difficult since one needs to have knowledge about colors, fashions, and styles among other factors. As a result, most people opt to use personal judgment, trial and error methods, or external assistance.

With the development of artificial intelligence and computer vision as well as fashion recommendation systems in recent times, there is a need to design technology-enabled intelligent models that can aid in making appropriate fashion decisions. Current fashion recommendation systems mostly consider the preferences of users, their physical appearance, seasons, shopping history among other aspects. Even though such approaches are useful, there is a drawback in such systems as they disregard environmental contexts, especially color themes of the event décor.

Recent advancements in the field of computer vision have shown that images can be processed to gain information on various features of interest such as color, texture, and pattern. Through the use of color clustering and segmentation, it is possible for computers to identify the dominant colors in a particular image. On the other hand, the fashion industry has also recognized the significance of color in determining fashion through the utilization of various color theories. This includes the complementary, analogous, and triadic color schemes among others.

However, the combination of event theme recognition and fashion is under-exploited due to the limited number of systems available today. Most existing systems are not able to process the images depicting the event decor, making it difficult for the computer system to make a suggestion according to the theme of the event. Another important factor is that some systems do not take into consideration the different events.

This paper seeks to overcome the shortcomings described above through the introduction of a novel artificial intelligence (AI)-based event theme color analysis and outfit recommendation system. Through this system, a user can upload an image of the event theme decoration and use the image processing technique to identify the dominant color scheme. Subsequently, the user will be asked to specify the nature of the event for the purpose of narrowing down the results.

In contrast to previous approaches to fashion recommendation, this one places an emphasis on contextual awareness by considering the color patterns found in the environment when suggesting

an appropriate clothing set. Not only does this mean that the suggested set will match visually; it also means that the recommendations will fit in terms of the chosen occasion.

In conclusion, this paper presents an effort to combine computer vision with the field of fashion recommendation by presenting a new system that analyzes event themes and makes appropriate suggestions based on those findings. This project adds value to the field of artificial intelligence in fashion assistance.

1.1 Background & Literature Survey

Artificial Intelligence integration in the fashion industry has received much interest, specifically in building intelligent fashion recommendation systems. A fashion recommendation system helps customers pick fashion products through several parameters, including user preference, physique, season, and history. The algorithm usually adopts machine learning algorithms, collaborative filtering, and content-based filtering methods to give suggestions. Despite the extensive use of these systems, few solutions consider environmental or contextual factors in the recommendation process.

Computer vision-based recommendation systems constitute another vital research area in fashion technology. Such algorithms employ computer vision models to analyze images of either clothing products or clients. For instance, CNNs have been extensively used in applications like clothing classification, style recognition, and outfit compatibility prediction. These studies indicate that visual features such as color, pattern, and texture contribute significantly to fashion compatibility assessment. However, these systems examine clothing products individually without considering how the outfit interacts with its visual context, such as event decorations.

Color Analysis and Theory is another promising line of investigation concerning fashion. Colors have an essential place in human perception and aesthetics. Principles of complementary, analogous, and triadic colors are discussed in the field of fashion design. Recently, researchers started applying clustering methods, such as K-means algorithm, in order to recognize dominant colors in photographs and thereby enable the automatic extraction of color schemes. Such approaches are extensively used in tasks related to picture retrieval, branding, interior design, etc., but still not employed in fashion recommendation, especially as regards environment-related recommendations.

Context-Aware Recommendation Systems are widely discussed in the fields of commerce, multimedia, etc. In those contexts, such systems use additional information, such as user location, activity, weather, and others, to improve recommendation quality. Several attempts have already been made in fashion to take advantage of context information, including weather conditions and seasons for example. Nonetheless, none of the existing approaches considers visual features of the surrounding environment in fashion recommendations.

Advancements in multimodal AI systems demonstrate the importance and possibilities that can be achieved using a combination of visual and contextual input in order to make more informed decisions. Using both image analysis and user interaction results in more precise and relevant recommendations. On the other hand, the use of multimodal AI in fashion does not involve analysis of environmental images but uses user images along with images from clothing databases.

While computer vision technology, color analysis techniques, and recommendations have been thoroughly explored in previous research, there is a noticeable gap in the field. Few researchers tried to design systems that would analyze images of events, extract colors from those and then use the obtained information for recommending appropriate clothing. There are very few systems that consider an event along with visual input while making recommendations, yet this element is very important when recommending an outfit for an event.

This research contributes to the current body of knowledge in the field of computer vision, color analysis, and recommendations systems, but introduces an innovative approach to solving problems in these areas by incorporating occasion and color harmony aspects into clothing recommendations.

1.2 Research gap

Innovations within the sphere of multi-modal AI prove the significance and opportunities that could be reached using the integration of both visual and contextual information in order to provide more accurate recommendations. The integration of both image processing techniques and user interactions provides more relevant and precise recommendations. At the same time, multimodal AI in the fashion industry doesn't utilize any environmental image processing but analyzes images created by the user and images from fashion databases.

Despite numerous studies in such fields as computer vision, color analysis, and recommendations, there is one significant area that lacks attention in this field of study. Almost none of previous

researchers attempted to develop an algorithm that would analyze events' images, extract color palette from it and use the collected information in order to recommend appropriate clothes for the given event. However, this aspect is very crucial when creating a recommendation regarding what clothes to wear for an event.

The proposed research represents an innovative approach towards solving problems associated with computer vision, color analysis, and recommendations by taking into account two crucial factors – event and color harmony.

First, previous work has focused mostly on the technical aspects of image retrieval and recommendation while neglecting other important factors such as cultural significance or user preferences.

Second, most existing approaches do not leverage the power of machine learning methods for intelligent image search or style recommendation. Instead, they rely on manual design of algorithms or rule-based systems.

Third, although extensive work has been done on the topic of color theory and harmony in design and fashion, these concepts have not been utilized by recommendation algorithms in an efficient manner. Most intelligent systems that do employ color theory do so on a superficial level, without using color matching and schemes in context.

Fourth, while current recommendation systems are aware of the contextual aspects of recommending clothing items, this awareness does not extend to the occasion-specific visual contexts related to the theme of an event.

Fifth, most existing solutions tackle only one problem at once, whether it is an issue of image-based color recognition or color harmony, rather than integrating multiple techniques into one system.

1.3. Research problem

The primary issue addressed by this study is the absence of an intelligent, context-sensitive fashion recommendation system which would be capable of analyzing images from the decoration of the event and recommending appropriate outfit colors and styles according to the thematic and visual elements as well as the kind of the event in question.

For instance, when participating in events like weddings, parties, office events, and any other thematic gatherings, people often encounter difficulties regarding choosing proper outfits which match their environment. In addition to the wearer's preferences, choosing an appropriate outfit also depends on the colors and aesthetics of the venue. However, current fashion recommendation

systems fail to offer adequate assistance for this type of situation since they mostly depend on user-centric data rather than on environmental factors.

So far, individuals have used subjective assessments, intuition, or advice from third parties to find a way to properly fit in with the chosen outfit. Moreover, there is currently no technological solution which could automate this process through analyzing an event decoration picture, determining its dominant colors and offering corresponding fashion suggestions accordingly.

To begin with, there are certain technical issues in this regard. The first one is that the images of event decorations need to be analyzed for extracting color schemes that are dominant. The second issue is mapping these colors to clothes colors. Thirdly, occasion-based constraints should be incorporated into the system to generate socially acceptable recommendations. Lastly, this whole system needs to be combined together into an application that provides real time suggestions.

1.4. Research Objectives

This study seeks to implement an Event Theme Color Analysis and Outfit Recommendation System that will help users choose appropriate outfits using event theme decoration pictures and event type. The intention here is to fill the existing gap between computer vision image analysis systems and fashion recommendation systems by incorporating color theory and context knowledge into one system.

1.4.1 Main Objective

The main objective of this research is:

A project to create and develop an AI-powered solution capable of analyzing images of event decorations, detecting the prevailing color scheme through computer vision algorithms, and offering appropriate suggestions regarding attire colors and styles compatible with the event occasion.

1.4.2 Specific Objectives

The specific objectives of the study are:

1. To gather a dataset consisting of images of event decorations covering various occasions, including wedding, party, corporate, and casual events.
2. To build a computer vision solution for extracting dominant colors from images of event decorations through image processing and clustering techniques.

3. To utilize the color theory principles, such as complementary, analogous, and triadic color schemes, for mapping the extracted colors to clothing colors.
4. To create a filtering algorithm based on the occasion chosen by the user to modify outfit recommendations based on the selected event type.
5. To establish recommendation logic for generating outfit recommendations for both male and female individuals using the extracted color themes and fashion guidelines.
6. To build a fully functional full-stack application for allowing users to upload their event images, choose their occasion, and get outfit recommendations instantly.
7. To test the developed application to ensure its efficiency in recommending visually suitable outfits for various occasions.

2. METHODOLOGY

2.1 Overview of the methodology

The current research utilizes an organized and systematic approach in designing and developing an AI-based Event Theme Color Analysis and Outfit Recommendation System. The methodology is specifically developed to solve the problem of recommending appropriate outfits based on the theme

of decorations at events and according to event context by employing computer vision, color science, and fashion recommendation.

Unlike the existing fashion recommendation models that consider only personal tastes of users or datasets of garments, the current study utilizes a context-based approach whereby recommendations are made after assessing the environment surrounding the event. This is why the methodology adopted in the current study is an end-to-end approach that converts event decoration images into useful fashion recommendations.

In this regard, the general procedure starts by obtaining images of event decorations that represent various occasions like weddings, parties, corporate events, etc. The images obtained are analyzed to extract dominant colors in the images. Based on the extracted colors, fashion recommendations are generated.

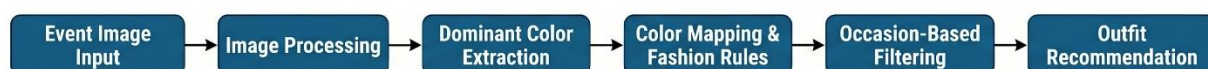
Further, once the color information has been extracted, color theory concepts such as complimentary colors, analogous colors, and triad colors are applied in identifying aesthetically matching clothing colors.

Besides the color-based analysis technique, the method incorporates an occasion-based filtering technique that enables different occasions to determine the styles and forms of suggested outfits. As such, in the case of more formal events like weddings or corporate events, more elegant outfits will be suggested, whereas casual events will yield less traditional recommendations.

Finally, the method involves the creation of a fully functional full stack application that enables the user to feed event images, select events and get outfit recommendations within seconds. It consists of a pipeline that includes a front-end interface for user interactions and a back-end pipeline for processing the inputs and generating outfit recommendations.

Generally, the methodology can be viewed as a multi-stage process involving the feeding of event images, color extraction from the image, color mapping, occasion filtering and finally recommendation generation.

Figure 2.1: Overall Research Methodology Workflow



2.2 Research design

The research in question is conducted in accordance with an applied research design with a considerable emphasis on system development and implementation. Applied research implies the presence of a real-life problem that should be addressed using specific methods. In this case, the problem revolves around difficulties faced by users when choosing outfits based on decoration themes at events and occasions. The current research does not aim at providing mere theoretical analyses; on the contrary, its purpose is to create an operational system capable of addressing practical challenges.

A particular research design employed in the process of completing the assignment may be described as design-oriented and development-based. According to this design type, the end product is expected to be a functional AI-based tool. The tool itself is created by following several stages, such as data collection, image processing, feature extraction, rule-based reasoning, and recommendation generation.

Technically speaking, the research design chosen for the project is data-driven. Event decoration images are regarded as the main data sources to be analyzed. Next, the system analyzes visual features of these images, with the emphasis being placed on the dominant color. Afterward, the collected visual data is converted into structured information and applied to generate recommendations.

Moreover, the research design utilizes a mixed-methods approach, which combines computer vision methods and fashion knowledge-based reasoning. While computer vision approaches allow for analyzing images and extracting dominant colors, fashion knowledge-based reasoning allows for applying the rules of color harmonization and filtering according to the type of event. As such, the system will be intelligent and transparent.

Furthermore, the study employs the prototype method to develop a full-stack software. In other words, the prototype will enable the user to interact with the system through uploading pictures of

events and obtaining relevant outfit suggestions. The use of the prototype method ensures that research outcomes will be applied in practice.

In conclusion, the research design embraces the principles of applied research, data analytics, and information systems development to create a context-aware fashion recommendation system.

2.3 System architecture

The architecture of the proposed Event Theme Color Analysis and Outfit Recommendation System takes the form of a pipeline where a particular event decoration image input is used in generating meaningful recommendations related to fashion. Such a multi-layered system design incorporates computer vision, color analysis, fashion rules mapping, and a user interface. The architecture of the system consists of the following four layers: Input Layer, Processing Layer, Recommendation Layer, and Application Layer.

1. Input Layer

The input layer is charged with capturing input from the user side into the system. The key input to the system is the event decoration image. This input serves as the visual indication of the occasion theme in cases such as weddings, parties, and corporate events among others. Apart from the image input, the user should also identify the type of occasion.

2. Processing Layer

The processing layer is the heart of the system and includes the analysis and extraction processes. After uploading an image of the event, the system will execute the following processes:

The image is pre-processed to ensure uniformity in terms of sizing and formatting.

An algorithm for dominant color extraction (e.g., K-Means Clustering) is run on the image to identify the key colors.

These colors are encoded to form a coherent palette reflecting the color theme of the event.

This ensures that the raw input image data is processed and turned into color-related information which can be used for fashion decisions.

3. Recommendation Layer

The recommendation layer generates outfit proposals by matching the extracted colors against the selected occasion in terms of visually pleasing colors and context-appropriate outfits. The processes performed in this layer include:

Using color theory techniques including complementary, analogous, and triadic color schemes to derive visually appealing outfits.

Rule-based mapping between the dominant colors of events and corresponding clothes.

Occasion-dependent filtering to account for different types of events such as formal, semi-formal, casual, etc.

4. Application Layer

The application layer provides the user interface and functionalities for interacting with the system.

The application layer is designed as a full-stack web application, whereby:

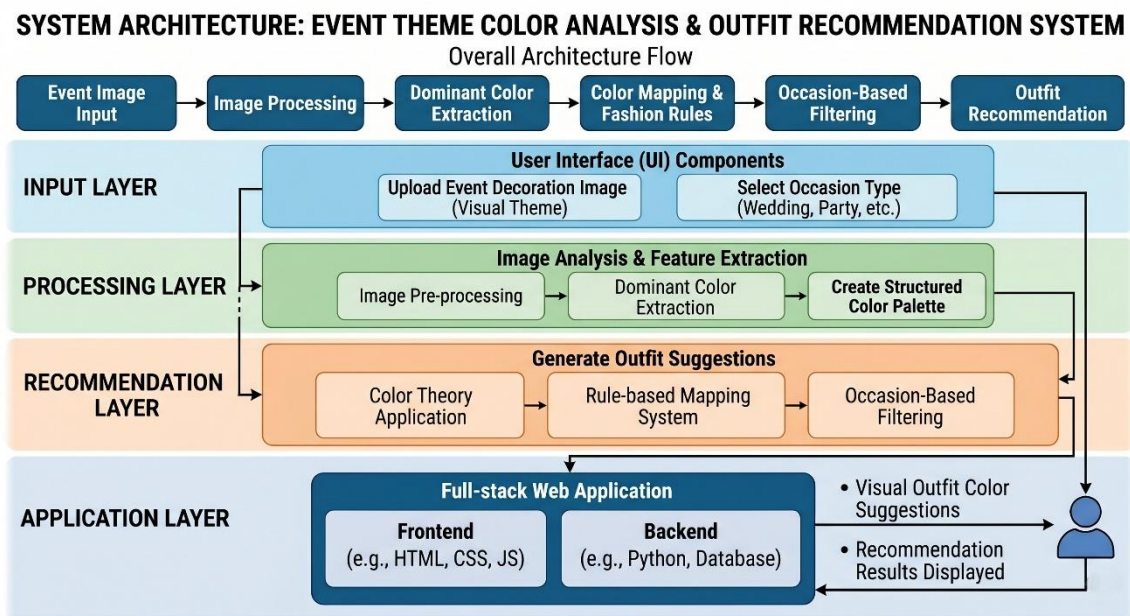
Users are able to upload pictures related to the event decorations.

Users choose the type of event that is to be celebrated.

The system recommends the color scheme and style of clothing to wear at the event.

The server processes the image and generates the recommendations.

This ensures visually compatible outfit choices for different occasions and genders.



2.4 Overall workflow of the proposed system

The flowchart below shows how the proposed Event Theme Color Analysis and Outfit Recommendation System works in its entirety, from taking the event decoration image input up to generating outfit recommendations based on the input image's dominant colors and the selected occasion type.

According to the proposed work flow, the process starts when the user inputs an image of the event decoration theme. The images represent the theme and color composition of an event environment, such as weddings, parties, corporate events, or casual gatherings. Aside from the image input, the user can choose the type of occasion. The choice of the occasion type will be a vital factor in recommending appropriate outfits.

After inputting the event decoration image and selecting the type of occasion, the system automatically proceeds with the image preprocessing process. The preprocessing stage involves resizing, normalizing, and other processes aimed at ensuring that the input image will be ready for analysis.

In the subsequent process, the system uses computer vision techniques to extract the dominant colors from the event decoration image. To do this, the system uses an algorithm such as K-Means Clustering to determine the colors in the event decoration image.

The next stage involves color analysis and mapping, whereby the colors that have been extracted in the previous phase undergo analysis using concepts in color theory. Suitable color matches are established by considering complementaries, analogous, and triadic colors. It is important to ensure that the outfits recommended by the algorithm match the event background.

The next step is occasion-based recommendations, whereby depending on the event chosen, specific fashion rules are applied to narrow down to particular outfit suggestions. In cases where the user chooses formal events, then outfit suggestions tend to be traditional and elegant. In cases of casual events, outfit recommendations can be more diverse and flexible.

The fourth phase includes generating outfit recommendations for both male and female clothing, which is done through the application of the colors extracted from the color palette, color theory, and occasion. Appropriate outfit color combinations are provided according to the selected event.

The last step entails presenting the results in the user interface level. The user interface provides the outfit suggestions in terms of style and color according to the selected event.

2.5 Data Collection

The importance of the data collection step cannot be overstated in the current research, since the effectiveness of the Event Theme Color Analysis and Outfit Recommendation System depends on how well the analysis will be conducted. Because this system is based on computer vision techniques, the choice of a good training set of images of event decoration is essential for the experiment.

Kaggle website, which is one of the most popular platforms offering an extensive collection of free image datasets for machine learning and data science projects, was selected for use in this project due to the wide range of freely available image data sets relevant to event decoration.

The collected dataset is comprised of a variety of images depicting decorations of various events, such as weddings, birthday parties, corporate events, celebrations, and casual parties, thus providing a sufficiently broad selection of examples for this research domain.

The next step was organizing and classifying the obtained images depending on the event type. It is necessary to organize the images according to specific occasions so that they can be easily assigned during the recommendation phase. For example, there may be a set of wedding decorations or a party. However, some datasets lack information about event types, requiring them to classify and sort manually.

After organizing and sorting the images, they were saved on the local machine and served as input for further image processing. The process of data preparation involved cleaning the data from

irrelevant images, duplicates, and other artifacts that would make the results unreliable and inconsistent.

The importance of this dataset is that it allows analyzing real-world event environments. The variability of the decoration styles and colors provides the basis for learning and extracting visual patterns needed for generating context-aware recommendations.

It is worth noting that the Kaggle dataset may require additional preprocessing to make it appropriate for the current study because it is not fully standardized for fashion purposes. Thus, data cleaning and classification have been applied to address this problem.

In conclusion, the Kaggle dataset contains a sufficient number of event decoration images for developing the proposed system.

2.6. Data Preprocessing and Structured Extraction

Following the acquisition of event decoration images from Kaggle, the next stage of the methodology entails preprocessing and feature extraction. This step is crucial since raw images cannot be analyzed or recommendations generated using the unprocessed images. As such, it is important to preprocess and transform the images into useful numerical and visual features before any analysis or recommendations can be made.

The preprocessing process starts with resizing the images into standardized dimensions, ensuring consistency in all the images in the dataset. Additionally, each image is transformed into the standardized color model (RGB) that will allow analysis of the colors used in the images.

At this point, the system eliminates noisy images, blurring, and images that do not contain event decorations.

One of the major steps in the feature extraction process within the scope of this research is the dominant color extraction. As the system is meant to work with event theme colors, finding the most representative colors within each image is crucial. Thus, a clustering algorithm such as the K-Means is used in order to identify the required amount of clusters within each image in the dataset.

In particular, each image is regarded as a set of pixels, where each of them has its own RGB coordinates within the RGB space. In this regard, the algorithm divides pixels based on their color into several clusters. Centroids of those clusters are considered dominant colors of the respective image. Together, these colors create the event decoration theme palette.

Once the dominant colors have been extracted from each image, they are structurized in order to use further in the recommendations generation process based on color harmony rules.

In addition, occasion-related features are prepared in the process of feature extraction. Each image in the dataset has its corresponding event label, such as 'wedding', 'party', 'corporate', and other event types. These features help to form the outfit recommendation output.

The resulting feature set is obtained by combining the characteristics related to colors and those related to occasions. In other words, the hybrid feature set will take into account not only visual attributes but also contextual attributes related to the event type.

On the whole, this process leads to the transformation of unstructured images into the required feature sets which are then used for intelligent outfit recommendations.

2.7. Data Cleaning and Preparation

However, even after the feature extraction stage, some steps needed to be taken to ensure consistency, accuracy, and appropriateness of data in the recommendation process. Although the event decoration images collected and color features extracted served as an excellent basis for the project, the raw data could have some flaws that would negatively impact the recommendation system. Thus, the stage of structured data cleaning and preparation followed.

Firstly, all irrelevant and of poor quality images were removed from the dataset. It was observed that certain images were of such low quality that they could not be utilized due to being over- or underexposed, blurry, etc., which meant that these images were not suitable for use as event decoration photos.

Then, to minimize potential errors in distribution of color, duplicates were found in the image dataset, and duplicates removal took place. This step is important because Kaggle datasets might have repetitions or high resemblance between images, which will lead to inaccurate data.

Finally, the extracted data related to the dominant color had to be processed as well. There were cases in which the extraction yielded inconsistent or even wrong results in terms of color, especially if there was low-lighting or complicated background.

After that, the data was arranged in a structured format where images and their dominant color palettes were matched with their corresponding occasions. Such a structured form of data would ensure that there is a proper correlation between the features and occasion type of each image.

Color values were also made consistent and normalized in order to standardize them in the database. The RGB values were standardized to ensure consistency among the colors and in certain instances converted to other types of color representation such as HEX code in order to make it easier to generate the color recommendation.

This phase also involved the verification of label consistency for the different occasions that had been used in the categorization of the images. In view of the fact that some of the images had been manually categorized depending on the nature of events, it was necessary to verify whether each picture has been correctly labeled to the right occasion such as wedding, party, corporate, or casual events.

Finally, the cleaned structured dataset was ready to be used in recommendation process and modeling.

2.8 Exploratory Data Analysis

Following the completion of data preparation, the next stage involved the exploration of the data through exploratory data analysis (EDA). In this stage, the researcher sought to gain a deeper understanding of patterns existing in the data. Information gained from this stage would be useful in coming up with appropriate recommendation logic.

The main objective of performing EDA in this study was to examine the distribution of dominant colors, the distribution of event categories, and the relationship between the two. As the recommendation system operates on visual color information, it was necessary to gain an understanding of color distributions across event categories.

The first analysis involved examining the distribution of event categories in the dataset. The data used in this study contains various types of occasions such as weddings, parties, corporate occasions, festive occasions, and casual occasions. The results of the distribution showed that certain event categories have more images than others. For instance, weddings and parties had many more images than other events.

Next, the analysis concentrated on dominant color schemes detected from images. It was noticed that the color palettes used for wedding occasions were light and elegant. They included colors like white, gold, pastel colors, and cream colors. On the other hand, colors that were commonly used

for party occasions included bright colors such as red, blue, purple, and neon colors. Color palettes for corporate events included neutral colors like gray, black, and navy blue.

Another aspect considered through EDA was the frequency distribution of different colors. Colors like white, gold, and red were very common in the images in the database. The information obtained from this observation will be useful since it affects the algorithm used by the system to recommend outfits.

Additionally, the relationship between colors and occasions was explored. It was realized that different occasions had unique visual identities depending on colors. This shows that considering occasion type is essential while recommending an outfit.

Variation in the nature of the lighting and the style of decoration in images is another significant observation made through exploratory data analysis. While some of the images had natural lighting, others used artificial lighting, which impacted color extraction accuracy.

The insights generated from exploratory data analysis were vital in giving information on the characteristics of the datasets and color distribution within them. They influenced the decision on the development of features and the recommendation algorithm.

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2.9 Feature Engineering

The role of feature engineering becomes very important in this research as it describes the method of mapping raw visual data and other contextual data into meaningful features for input to the recommendation system. The reason being that the recommendation system under investigation uses an event decoration image analysis approach and therefore feature engineering is done using color-based and occasion-based features.

Feature engineering in this research comprises three major elements which are color-based features, color harmony features, and occasion-based features.

Structured Color Features

The first feature set to be extracted is that of the structured colors from the event decoration pictures. The image processing and clustering methods, such as K-Means algorithm, enable extracting the dominant colors of each image, which will then be mapped to the standard formats such as RGB and HEX representations.

Thus, each image will be mapped to a set of structured color feature vectors that reflect its visual theme, which in turn reflects the visual theme of the event decoration itself. These color features are the key for the recommendation process, since they will guide the choice of colors of the outfit.

Color Harmony Features

Besides the direct extraction of color information, the use of color theory will be utilized to improve the performance of the model. Additional color harmony features will be extracted from the dominant colors of the image according to the following set of rules:

Complementary Colors: Two colors that are located on the opposite sides of the color wheel and provide contrast.

Analogous Colors: Three colors that are placed in close proximity on the color wheel and create a harmony.

Triadic Colors: Three colors located evenly across the color wheel providing visual contrast.

Occasion Based Features

Since the system is aware of its context, occasion type is also one of the important features. Event images have occasion tags like weddings, parties, corporate events, and casual meetings.

These occasion tags are then turned into categorical features which affect the final outcome. For

This feature guarantees that the outfit suggestions are socially appropriate.

Hybrid Feature Representation

The resulting feature representation that this system uses is a blend of:

Extracted dominant color features

Derived color harmony features

Contextual occasion features

With this hybrid combination of the three aforementioned features, this system can process visual content (color data) and contextual data (occasion type) effectively.